TEACHER GUIDE

ANSWERS AND MARKS

GRADE 10 · INFORMATION AND COMMUNICATION TECHNOLOGY

UNIT 9
Database



! FOR THE TEACHER ONLY

This booklet has the answers to every exercise in the Unit 9 Student Workbook, in the same order. The exercise letters match the workbook.

i WHAT IS IN THIS GUIDE

- **Marks and pacing** - every exercise and its marks
- **Class marks record** - one page to copy for each class
- **Practicals at a glance** - Practicals 1 - 5, materials, a quiz, a practical test
- **Answers** - chapters 1 to 4, in the same order as the workbook
- **Unit 9 Test** - the marking scheme and the common mistakes
- **Extra** - textbook links, homework ideas and teaching tips

★ BEFORE THE FIRST LESSON

- ☐ MS Access 2010 (or LibreOffice Base) opens on every computer.
- ☐ A database with the Student and Marks tables is ready for Practicals 4 and 5.
- ☐ Every student has a folder to save his or her database.

"Students who build the tables themselves never forget the keys."

- THE KING OF ICT

TEACHER GUIDE

Marks and Pacing

You decide how much class time each part needs

Chapter	Exercise	Marks
Chapter 1	A Manual or electronic?	8
	B Find the problem	4
	C True or false	6
	D Match the features	6
	E Field properties	6
Chapter 2	A Which key is it?	6
	B Data duplication	6
	C Label the table	6
	D Degree and cardinality	8
	E Choose the correct answer	6
Chapter 3	A Choose the data type	7
	B Which data type?	8
	C Which relationship?	6
	D Complete the Student_Sport Table	7
	E Count the related records	6
	F Draw the relationships	6
Chapter 4	A Label the navigation bar	7
	B Run the query	6
	C Calculate the totals	8
	D Table, query, form or report?	8
	E The report total	4
	F Answer these	4
	G Crossword	10
End	Unit 9 Test	40
TOTAL		**189**

! SUGGESTED SHAPE OF ONE CHAPTER

READ	Teach the chapter pages. Students read with you and underline the key words.
DO	The practical tasks on real computers, in pairs.
WRITE	The written exercises. Walk around and help.
MARK	Mark the work together. Fill the tick boxes.

★ TEACHING TIP

- Start with a real register: the class attendance book is a manual database. Ask how it could be electronic.

TEACHER GUIDE

Class Marks Record

Copy this page for each class

No	Student name	Ch 1 (30)	Ch 2 (32)	Ch 3 (40)	Ch 4 (47)	Test (40)	Total (189)
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							
11							
12							
13							
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19							
20							
21							
22							
23							
24							
25							
26							
27							
28							
29							

TEACHER GUIDE

Practicals at a Glance

Textbook activities and steps on pages 273 - 311

Practical	Students do	Check that ...
1 (pp. 273 - 274)	the field properties of Fees: Currency, 2 decimal places, default 1000, validation rule >=1000, required	800 is refused, 2500 is accepted and an empty fee is refused - this is validity.
2 (p. 278)	follow the records of Table 9.4 from table to table	Firosa (101): 2222 Folk Songs, 1111 Basic Maths Concept. Manoj (100): 3333 Drama, 4444 Principles of Media.
3 (pp. 285, 287 - 289)	a list of 10 library books; the Library table, its key and its data	Accession_Number has the key icon, Price is Currency, Number_of_books is Number.
4 (pp. 297 - 299)	the Student and Marks tables joined one-to-many with referential integrity	the 1 and many signs show; marks cannot be entered for a student who is not in the Student table.
5 (pp. 300 - 311)	a form with a button, a query with the Criteria 2014 and a Total column, a report with a total	the 2014 query shows 7 records; the 2013 totals are 190, 212, 126, 227, 161, 142, 224; the report total is Rs. 3,920.00.

SAFETY AND MATERIALS

- Computers with MS Access 2010 (or LibreOffice Base). Check that it opens before the lesson.
- Prepare a database with the Student and Marks tables (Figures 9.13, 9.14) for Practicals 4 and 5.
- Every student saves in his or her own folder. Never type real phone or ID card numbers of students.
- A printer for Practical 5 - or let students stop at Print Preview.

Textbook links

Workbook	Textbook pages	Activities
Chapter 1 - Databases and their features	269 - 275	-
Chapter 2 - Fields, records and keys	275 - 283	278
Chapter 3 - Creating a relational database	284 - 299	285
Chapter 4 - Forms, queries and reports	300 - 311	311 (questions)
Unit 9 in One Page	312 (summary)	-

TEACHING TIPS

- Draw a table on the board and point: a column is a field, a row is a record. Repeat it every lesson.
- Teach "degree = columns, cardinality = rows". Remind the class that the heading row is not a record.
- Ask students to write the result of a query on paper first. Then run it and compare.

? QUICK ORAL QUIZ - START OR END A LESSON WITH IT

Ask the class ...	Answer
1. A collection of logically organised data is a ...?	**database**
2. The number of fields in a table is its ...?	**degree**
3. The number of records in a table is its ...?	**cardinality**
4. Two special features of a primary key?	**not null, no duplicates**
5. The primary key of the Sports Table (Table 9.10)?	**Year + Sport**
6. The best data type for a telephone number?	**Text**
7. The four objects of a DBMS?	**tables, queries, forms, reports**
8. Students and sports: which relationship?	**many-to-many**
9. Which object gives a printed copy?	**a report**
10. In which row of the query grid is 2014 typed?	**Criteria**

> PRACTICAL TEST - 30 MINUTES, 20 MARKS (OPTIONAL)

Use the computers of Practicals 3 - 5. Data types for step 1: Text, Text, Date/Time, Text, Currency.

Step	The student must ...	Marks
1	Create a database with your name. Make the **Student** table of Figure 9.13 with the right data types.	5
2	Make Admission_Number the primary key. Enter three students.	3
3	Make the **Marks** table. Relate it to the Student table with Enforce Referential Integrity.	4
4	Make a form for the Student table with an Add New Record button.	3
5	Make a query that shows only the marks of 2014 (Criteria 2014).	3
6	Save and close the database.	2
	Total	**20**

★ TEACHING TIPS

- Before choosing a primary key, ask: can this column repeat? Can it be empty? Two "no" answers make a key.
- Make students underline the primary key in every table they draw. It is a habit the O/L examiners reward.
- Use the students' own admission numbers to show why a name is never a good primary key.
- For a composite key ask: which two columns together never repeat? Try Year + Sport on the board.
- Show data duplication with Table 9.15: it looks like five students scored above 60, but there are only three.

ANSWERS

Chapter 1

Databases and their features

A Manual or electronic?

8 marks

The statement	Manual	Electronic
Data can be validated.	☐	☑
More man power is needed.	☑	☐
A very small space is needed to save data.	☐	☑
Difficult to delete unnecessary data.	☑	☐
The data input order can be changed.	☐	☑
Access control is difficult.	☑	☐
Easy to share data.	☐	☑
Less credible.	☑	☐

1 mark each. All the statements come from Table 9.1 (page 271).

B Find the problem

4 marks

1. **Firosa Lai and Manoj Dayarathne**
2. **2**
3. **data inconsistency**
4. **1**

1 mark each. In Table 9.4 the name is kept only in the Student table, so it is changed once.

C True or false

6 marks

1. **T - A data set is a collection of data that is not logically organised.**
2. **F - A manual database is more efficient than an electronic database.**
3. **T - In a manual database it is difficult to update data.**
4. **F - A good database has a lot of data redundancy.**
5. **T - The relational model is one of the database models.**
6. **F - Encrypted data (cipher text) can be read easily by anyone.**

D Match the features

6 marks

1. Minimal data redundancy → B	**2.** Data consistency → E
3. Increased efficiency → D	**4.** Increased accuracy → F
5. Increased validity → A	**6.** Security → C

E Field properties

6 marks

1. Field name: **Fees**	**2.** Data type: **Currency**	**3.** Decimal places: **2**
4. Rs. 800: **No**	**5.** Rs. 1000: **Yes**	**6.** Rs. 2500: **Yes**

1 mark each. The Validation Rule >=1000 refuses 800; 1000 and 2500 are accepted.

ANSWERS

Chapter 2

Fields, records and keys

A Which key is it?

6 marks

The field (or fields)	Primary key	Composite primary key	Foreign key
Accession No (Book Table 9.8)	☑	☐	☐
Year + Sport (Sports Table 9.10)	☐	☑	☐
Registration No (Subject Table 9.13)	☐	☐	☑
Candidate No (Subject Table 9.13)	☑	☐	☐
Admission No (Marks Table 9.17)	☐	☐	☑
Year + Term + Admission No (Marks Table 9.17)	☐	☑	☐

B Data duplication

6 marks

1. 5	**2. 3**	**3. 2**

4. Any three of (1 mark each): A primary key cannot be selected to keep the identity of records. Data cannot be analysed properly. Efficiency is reduced. There is a high chance of inaccurate data (Malikkar may become Malakkar). It is difficult to input data. Errors can happen when data is deleted. It is difficult to update data.

C Label the table

6 marks

1. Field name	**2. Field (column)**	**3. Record (row)**
4. Primary key	**5.** Degree: **3**	**6.** Cardinality: **3**

1 mark each. The table is Table 9.16 (Student Table): 3 fields and 3 records. Accept "field" and "record" without the words in brackets.

TEACHING TIP

- Let two groups act as the Student Table and the Marks Table, and join them with admission numbers.

D Degree and cardinality

8 marks

Table	Degree	Cardinality
Table 9.2 - Admission_table	**5**	**2**
Table 9.3 - Library_membership_table	**4**	**4**
Table 9.8 - Book Table	**5**	**7**
Table 9.10 - Sports Table	**4**	**6**

1 mark for each number. The heading row is not counted.

E Choose the correct answer

6 marks

1. (a) 6	**2.** (c) 7
3. (c) Admission No	**4.** (c) unique
5. (c) Year + Sport	**6.** (a) duplication

★ TEACHING TIPS

- Draw the arrow from the foreign key to the primary key every time (textbook page 290). Say it aloud.
- Remind students that a telephone number is Text: it is never used for calculations, and it starts with 0.
- Show both MS Access and LibreOffice Base if you can. The ideas are the same; only the menus change.
- Prepare a ready database with the Student and Marks tables. Students then spend the time on queries.
- Print one report for the class notice board. Students like to see their work on paper.

ANSWERS

Chapter 3

Creating a relational database

A Choose the data type

7 marks

1. Admission Number: **Text**	**2.** Date of Admission: **Date/Time**
3. Subject Code: **Text**	**4.** Doctor's Fee: **Currency**
5. Department Name: **Text**	**6.** Number of Passengers: **Number (Integer)**
7. Is Registered?: **Yes/No (Boolean)**	

1 mark each. Admission Number and Subject Code are Text, as in Figure 9.13: they are never used for calculations.

B Which data type?

8 marks

Value	Integer	Real	Text	Currency	Date/ Time	Boolean
-23	☑	☐	☐	☐	☐	☐
8.125	☐	☑	☐	☐	☐	☐
Rs.35.00	☐	☐	☐	☑	☐	☐
7.35 a.m.	☐	☐	☐	☐	☑	☐
True	☐	☐	☐	☐	☐	☑
NIC number 889534731V	☐	☐	☑	☐	☐	☐
Telephone no. 0112785123	☐	☐	☑	☐	☐	☐
-2.64	☐	☑	☐	☐	☐	☐

C Which relationship?

6 marks

The two things	One-to-one	One-to-many	Many-to-many
A student and his or her scholarship Index No	☑	☐	☐
A student and the fee receipts he or she gets	☐	☑	☐
Students and the sports they play	☐	☐	☑
A student and his or her term marks	☐	☑	☐
A citizen and his or her NIC number	☑	☐	☐
Students and the library books they borrow	☐	☐	☑

1 mark each. A student has one scholarship index number and one NIC number; a student gets many receipts and many marks records; students and sports, students and library books are many-to-many.

D Complete the Student_Sport Table

7 marks

Student_Sport Table

Admission No	Sport No	Team
1426	S001	A
1426	S002	A
1428	S001	B
1428	S002	A
1428	S003	B
1429	S001	A
1429	S003	A

1 mark for each row (Table 9.30, page 295). The textbook prints the 6th row as "1429, 001, A" - the right Sport No is S001. Accept the rows in any order.

★ TEACHING TIP

- Never let students type real phone or ID card numbers of their friends. Use the textbook data.

E Count the related records

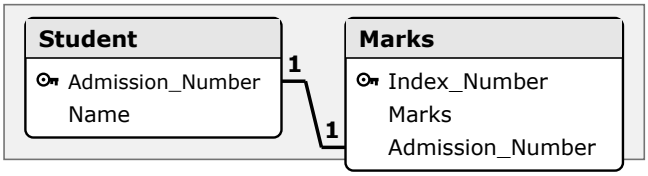
6 marks

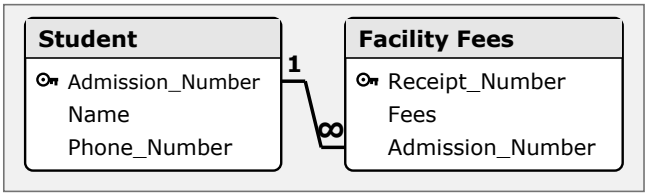
1. 2	**2. 0**	**3. Rs. 120.00**
4. 3	**5. 3**	**6. B**

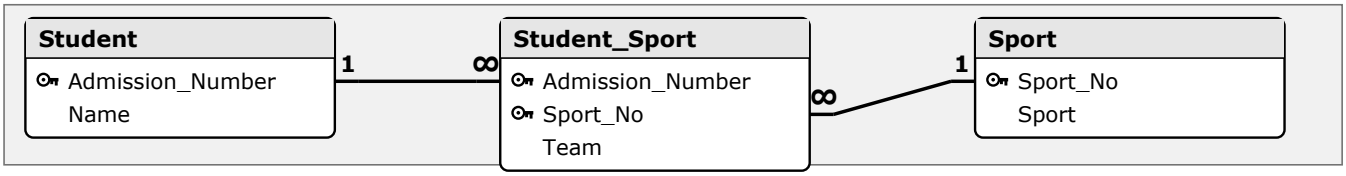
1 mark each. Student 1426 paid twice (40.00 + 80.00); 1427 has no receipt.

F Draw the relationships

6 marks







1 mark for each correct line, 1 mark for the 1 and many signs. The many sign is always next to the table with the foreign key.

ANSWERS

Chapter 4

Forms, queries and reports

A Label the navigation bar

7 marks

1. First Record	**2.** Previous Record	**3.** Existing (current) Record	**4.** Total Records
5. Next Record	**6.** Last Record	**7.** New (empty) Record	

1 mark each (Figure 9.20). Accept "current record" for 3 and "total number of records" for 4.

B Run the query

6 marks

The query result

Name	Subject	Marks
Pubudu	**Maths**	**67**
Faris	**English**	**89**
Kamalan	**Science**	**75**

2 marks for each row (Figure 9.25). Pubudu's English mark (50) is not more than 60, so that record is not shown.

TIP: Access saves the records by itself, but a changed table design must be saved. Remind them often.

C Calculate the totals

8 marks

Surname_with_Initials	Maths	Science	English	Total
Karunajeewa R.J.C.	67	45	78	**190**
Somarathna A.B.G.	76	81	55	**212**
Rasaputhram S.N.	37	54	35	**126**
Jayasena L.K.R.	87	72	68	**227**
Malikkar M.N.M.	62	56	43	**161**
Radhakrisna T.U.	32	45	65	**142**
Faslina M.G.	70	89	65	**224**

1 mark for each total (Figure 9.33). Highest total: **Jayasena L.K.R.** (227) - 1 mark.

D Table, query, form or report?

8 marks

The job	Table	Query	Form	Report
Stores the data in rows and columns.	☑	☐	☐	☐
Shows only the marks of 2014.	☐	☑	☐	☐
A screen to enter one record at a time.	☐	☐	☑	☐
Prints the marks for the notice board.	☐	☐	☐	☑
Calculates a Total column for each student.	☐	☑	☐	☐
Has a Record Navigation Bar.	☐	☐	☑	☐
Shows the total of the fees in its footer.	☐	☐	☐	☑
Is planned with field names and data types.	☑	☐	☐	☐

E The report total

4 marks

1. **In the Report Footer**
2. **No field is given to it**
3. **=Sum([Student Table]![Admission Fees])**
4. **Rs. 3,920.00**

1 mark each (textbook pages 308 - 310): 7 x Rs. 560.00 = Rs. 3,920.00.

F Answer these

4 marks

1. What is the difference between a form and a report? (2 marks)

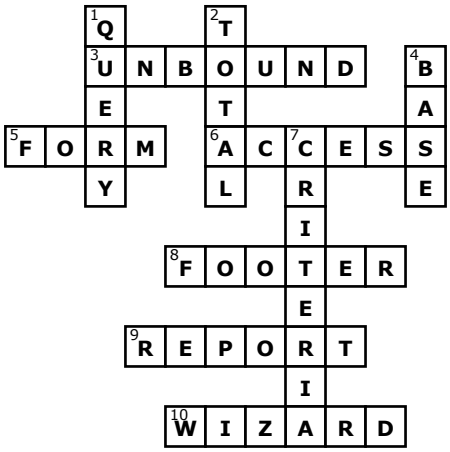
A form is used to enter data into a table and to display it on the screen (1 mark). A report presents information as a printed copy (1 mark).

2. Write the expression of a Total column that adds the Maths, Science and English marks. (2 marks)

Total: [Marks]![Maths]+[Marks]![Science]+[Marks]![English] (1 mark for the column name Total:, 1 mark for the three fields joined with +).

G Crossword

10 marks



Across: 3 UNBOUND, 5 FORM, 6 ACCESS, 8 FOOTER, 9 REPORT, 10 WIZARD

Down: 1 QUERY, 2 TOTAL, 4 BASE, 7 CRITERIA

MARKING SCHEME

Unit 9 Test

Total 40 marks

1 Choose the correct answer

8 marks (1 each)

1. (b) database	**2.** (c) Base
3. (b) cardinality	**4.** (c) Currency
5. (d) Report	**6.** (c) many-to-many
7. (c) table	**8.** (a) A and B only

2 Fill in the blanks

8 marks (1 each)

1. inconsistency	**2.** encryption	**3.** field	**4.** composite
5. Boolean	**6.** Design	**7.** Criteria	**8.** Builder

3 Look at the tables

8 marks (1 each)

1. Field name	**2. Record**
3. Primary key	**4. Foreign key**
5. The degree of the Facility Fees Table: **4**	**6.** The cardinality of the Students Table: **3**
7. The relationship between the two tables: **one-to-many**	**8.** Payments of 1426: **2**

1 = a field name (Name), 2 = a record, 3 = the primary key of the Facility Fees Table, 4 = the foreign key (Admission No). The Facility Fees Table has 4 fields; the Students Table has 3 records.

4 Answer these

16 marks

1. What is a primary key? Write two of its special attributes. (3 marks)

A column (or a combination of columns) which identifies a record uniquely (1 mark). It cannot be empty - not null (1 mark). It cannot have duplicate values (1 mark).

2. Write the most suitable data type for these fields of the Library table: Title, Price, Number of Books, Registration status. (4 marks)

Title - Text, Price - Currency, Number of Books - Number, Registration status - Yes/No (Boolean). 1 mark each.

3. Students and sports have a many-to-many relationship. Write the field names of the three tables that are needed. Underline the primary key of each table. (3 marks)

Student (Admission No, Name, Date of Birth); Sports (Sport No, Sport, Number of permanent members); Student_Sport (Admission No, Sport No, Team). 1 mark for each table with its key.

4. Write two differences between a manual and an electronic database. (2 marks)

Any two rows of Table 9.1, e.g. an electronic database is more efficient and more accurate; it is easy to update, share and validate data; it needs very little space. 1 mark each.

5. What are forms used for? What are reports used for? (2 marks)

Forms: to enter data into a table and to display the data (1 mark). Reports: to get printed copies of important information (1 mark).

6. Write the expression for the total of the fees in the Report Footer. (2 marks)

=Sum([Student Table]![Admission Fees]) (1 mark for =Sum( ), 1 mark for [Student Table]![Admission Fees]).

WHAT STUDENTS USUALLY GET WRONG

- They count the heading row as a record. The cardinality counts only the data rows.
- They choose Name as the primary key. Names can be repeated; an Admission No cannot.
- They draw the arrow from the primary key to the foreign key. It goes **from the foreign key**.
- They store telephone numbers as Number. Telephone numbers are Text - the 0 at the start is lost.
- They put the many sign on the wrong side. It is next to the table with the foreign key.
- For many-to-many they add a foreign key to one table only. A **new table** with both keys is needed.
- They type 2014 in the Field row of a query. It goes in the **Criteria** row of the Year column.

HOMEWORK AND EXTRA PRACTICE

- Textbook Activity, page 311: questions 1 - 12, two questions a day.
- Activity, page 285: a manual table of ten books at home, with a data type for every field.
- Draw the three tables of students and the sports they play. Underline the keys and draw the arrows.
- Fast learners: make a query of the 2013 Science marks above 60 and print a report of it.

TEXTBOOK MISPRINTS - THE WORKBOOK USES THE RIGHT VALUES

- Page 277: Book Number 3333 is printed twice; the 4th row is 4444 (Principles of Media), as in Table 9.4.
- Page 281: Table 9.12 gives the dates of birth as 2014; Table 9.2 has 2/9/2008 and 4/2/2008.
- Pages 289 - 290: Meenadevi has 186 marks in Table 9.19 but 181 in Table 9.20 (the workbook uses 186).
- Pages 289 - 291: two sections are numbered 9.4.2 (one-to-one and one-to-many).
- Page 294: in Table 9.27 the Name column is headed "Facility fees Table".
- Page 295: Table 9.30 prints "001" as the Sport No of student 1429; it is S001.
- Page 286: "Oracle Cooperation" is Oracle Corporation. Page 287: "Auther" is Author.
- Page 303: the student A2 is "Fairs" in the Students table and "Faris" in the result (the workbook uses Faris).

TEACHING TIPS

- The many sign always stands next to the table with the foreign key. Check it in every diagram.
- For many-to-many, draw the new table in the middle. Its key is made of the two foreign keys.
- Turn on Enforce Referential Integrity, then try to enter marks for a student who does not exist.
- Show the + mark in front of a student in the Student table. Students enjoy entering marks through it.
- A form shows one record, a table shows many. Let students switch between Form View and Datasheet View.
- Play "navigation bar": call out first, previous, next, last or new, and let pairs click quickly.
- A criteria must be typed under the right field. Check this first when a query shows nothing.